Edge AI & TinyML: Hardware and Ecosystem

From Models → Real Devices

- Understanding hardware platforms
- Development tools & setup
- First embedded program

Why Hardware Matters

Hardware enables AI models to run locally with limited power, memory, and computational resources.

- Severe Resource Constraints
- Power and Thermal Efficiency
- Real-Time Latency
- Data Privacy and Security
- Performance Optimization

Specialized hardware, such as Microcontrollers (MCUs) and Neural Processing Units (NPUs), delivers low-latency, real-time inference, high energy efficiency (25–300 mW), and enhanced data privacy, making "always-on" intelligence feasible.

Hardware-Software Co-design: Successful deployment requires matching the model architecture to the hardware, using techniques like quantization (reducing precision) and pruning to make models fit within hardware constraints

Hardware Evolution

From General-Purpose to Specialized Computing

- Traditional CPUs designed for broad tasks
- Edge AI requires hardware optimized for neural networks

Edge Hardware Spectrum

- Ultra-low-power microcontrollers for sensors & wearables
- Edge computers & AI accelerators for vision & robotics

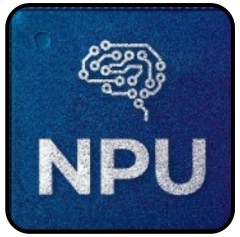
Purpose-Built AI Hardware

- Accelerates neural network processing locally
- Designed for power efficiency & real-time performance

Specialized hardware enables intelligent processing directly at the edge & MCU

Key Hardware Technologies for Edge AI / TinyML

Edge AI & Tiny ML performance depends on choosing the right processing unit. Different hardware balances performance & power based on device needs.

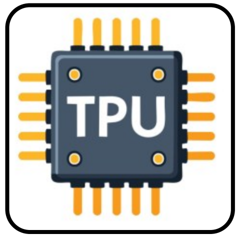


Neural Processing Units (NPUs)

- Optimized for on device edge AI
- Unlike CPUs/GPUs, NPUs speed up AI operations while keeping latency and power consumption very low.
- Examples: Arm Ethos, Apple Neural Engine

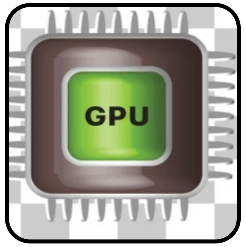
Vision Processing Units (VPUs)

- Designed for computer vision & image processing
- Ideal for cameras & surveillance
- Example: Intel Movidius Myriad X



Tensor Processing Units (TPUs)

- ASICs built for machine learning acceleration
- Compact & high-performance inference
- Example: Google Coral Edge TPU

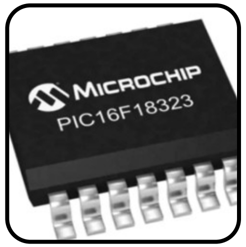


Graphics Processing Units (GPUs)

- High-performance parallel processing.
- Excel at heavy computing like AI model training (CNNs, LLMs), Robotics & advanced vision workloads
- High power consumption and cost
- Example: NVIDIA Jetson series

Field-Programmable Gate Arrays (FPGAs)

- Reconfigurable hardware for custom AI tasks
- Flexible & high-performance
- Vendors: AMD (Xilinx), Intel

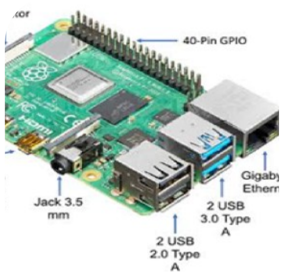


Microcontrollers (MCUs)

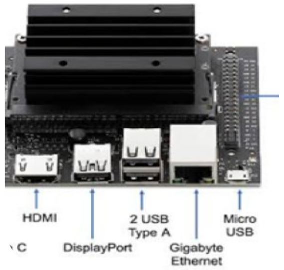
- Small computer on a chip, Includes CPU, memory, I/O. Used in embedded systems
- Ultra Low power with limited resources
- Tiny ML applications like Wake-word detection & sensor analysis
- Example: STM32 series

Edge AI / Tiny ML Hardware Example

Edge Computers (High Capability) - Run Linux & advanced AI models. Used in gateways, smart cameras & industrial edge devices



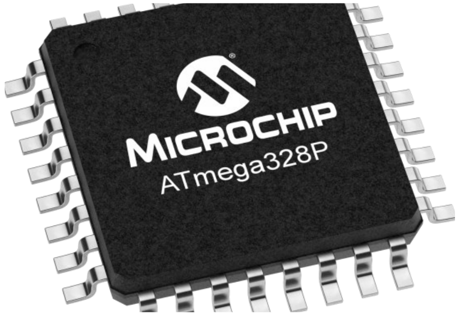
Raspberry Pi 4 Model B, a popular general-purpose credit-card-sized computer featuring a GPIO header, USB ports, and Ethernet connectivity.



NVIDIA Jetson Nano Developer Kit, designed specifically for running AI applications and neural networks, featuring a powerful GPU.

Microcontrollers (Ultra-Efficient)

- MCUs with limited compute & memory. Extremely low power consumption. Used in sensors, wearables & battery-powered devices



Microchip ATmega328P microcontroller. Frequently used as the processor in Arduino boards, such as the Arduino Uno

Key Ecosystem Characteristics for Edge AI / Tiny ML

Hardware-Software Co-Design

Efficiency is achieved by designing software that directly utilizes specialized hardware instructions, such as ARM's **CMSIS-NN** kernels, which maximize performance on Cortex-M processors.

Edge-Cloud Collaboration

While the **inference** (decision) happens locally, the **training** and large-scale model management often still reside in the cloud (e.g., Azure IoT Edge, AWS SageMaker).

Privacy & Speed

By processing data on-device, this hybrid model ensures data sovereignty and near-instant decision-making, which is critical for applications like **autonomous vehicles** or medical wearables.

Ultra-Low Power

The ecosystem is designed so that devices like industrial sensors can run on milliwatts of power, functioning for years on a single battery.

Hardware-Software Co-Design Example

Application Layer
- ML Model

Framework Layer
- TensorFlow Lite

Firmware Layer
- Embedded C/C++

Hardware Layer
- MCU (ESP32 / Arduino)

Evolution of Decision-Making Systems